Sheet 71 Mon. 24 May 2027 Weighted trees: multiplying along a path

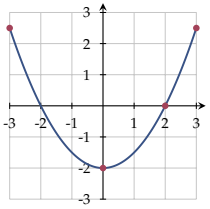
1. Warm-up · Flash Calculate $0.3 \times 0{,}9$.
2. Core · Method With the tree from sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.
3. Core + · Method Deduce $P(T)$.
4. Back · Factorise Factorise $x^2 - 9$.
5. Running thread · Flash The side $[BC]$ of length 13 cm is the hypotenuse of a right-angled triangle ABC. What is the radius of its circumcircle?

Sheet 72 Tue. 25 May 2027 From the table to the tree and to symbolic notation

1. Warm-up · Flash Express $\frac{30}{60}$ as a probability.
2. Core · Method From the table on sheet 69, build the tree (F or G, then S or $\overline{S}$): give $P(F)$, $P_F(S)$ and $P_G(S)$.
3. Core + · Translate Write symbolically and calculate "the probability of being a boy who does not do sport".
4. Back · Flash A car travels at constant speed: 150 km in 2 h. What distance in 5 h?
5. Running thread · Statement⟶ formula A taxi charges a 2,000 Ar pick-up fee then 1,500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

Sheet 73 Thu. 27 May 2027 Reversing the conditioning: P$_A(B)$ and $P_B(A)$

1. Warm-up · Flash What is the difference between $P_M(T)$ and $P_T(M)$?
2. Core · Method With the data from sheet 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.
3. Core + · Method Rare disease (1% of people ill); test positive for 99% of the ill and 5% of the not ill. Calculate $P_T(M)$.
4. Back · Graph Curve of f :



Read $f(2)$ and the preimages of 2.5.

5. Running thread · Method Procedure: choose a number; multiply it by -2; add 10; divide by 2. Result for 7? Expression in x ?

Sheet 74 Mon. 31 May 2027 False positives and false negatives

1. Warm-up · Flash Define "false positive" and "false negative".
2. Core · Method 10,000 people, 2% ill; the test is positive for 90% of the ill and 10% of those who are not ill. How many false negatives, false positives and positives in total?
3. Core + · Interpret With the previous data, calculate the proportion of people truly ill among the positives and comment.
4. Back · True/false True or false? Justify. The area of a square is proportional to the length of its side.
5. Running thread · Refute Conjecture: "for every integer n, $n^2 + n + 41$ is prime". Test $n = 40$.

Consolidation, fluency and transition to Première

Sheet 75 Tue. 1st June 2027 Synthesis problems drawing on several areas

1. Warm-up · Flash Solve 3 $(x-1) = x+5$.

2. Core · Method For which value of k are points $A(1;1)$, $B(3;5)$ and $C(k;9)$ collinear?

3. Core + · Method A price rises by 10% then falls by 10%. What is the overall change?

4. Back · Flash Total surface area of a cuboid with dimensions 4, 3, 2 (cm)?

5. Running thread · Flash Calculate the exact volume of a sphere with radius 3 cm.

Sheet 76 Thu. 3 June 2027 Review of the six skills

1. Warm-up · Metacognition State the six mathematical skills developed in lycée.

2. Core · Metacognition Match: "express a statement as an equation"; "draw a curve"; "justify an equality by a written calculation" to a skill.

3. Core + · Metacognition To solve 2 $x+3=11$, which skill is mainly used?

4. Back · Problem Water at 20 °C heats up by 3 °C per minute: $T(t) = 20+3t$. After how many minutes does it reach 65 °C?

5. Running thread · Flash Solve in $\mathbb{R}$: $0{,}5x+1{,}2=2{,}7$.

Sheet 77 Mon. 7 June 2027 Targeted revisiting of prerequisites for Première

1. Warm-up · Flash Expand $(x+3)^2$.

2. Core · Method Check that $x^2-5x+6=(x-2)(x-3)$ and deduce the solutions of $x^2-5x+6=0$.

3. Core + · Anticipation For $f(x)=x^2$, calculate the rate of change $\dfrac{f(3)-f(1)}{3-1}$.

4. Back · Construction steps We place M and N midpts of $[AB]$ and $[AC]$ then we draw $[MN]$. What can be said about (MN) and (BC) ? Calculate MN if $BC = 9$ cm.

5. Running thread · Write up Prove that a triangle with sides 7, 24 and 25 is right-angled.

Sheet 78 Tue. 8 June 2027 Synthesis problems: choosing and justifying a strategy

1. Warm-up · Code reading Output of $f(2)$?

```
def f(x):
    return x**2 - 3*x + 1
```

2. Core · Completion Complete so that the function returns True when A, B, C are collinear:

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == ....
```

3. Core + · Completion Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        return x
    else:
        return ....
```

4. Back · Flash Write in decimal notation 2.5 $\times 10^3$ then 6.2 $\times 10^{-2}$.

5. Running thread · Flash Give the image of $A(2;-3)$ about the origin.